

BUILD GUIDE · ORIGINAL DESIGN · SMALL MODEL

Adjustable Therapy Bench

Version 2 — Slot Channels + Padded Top

5–9"

HEIGHT RANGE

1"

INCREMENTS

20×10"

TOP SURFACE

~\$50

MATERIALS EST.

3–4 hrs

BUILD TIME

Built for a son with Cerebral Palsy. Designed for floor sitting work and home therapy practice. One sheet of plywood, around \$50 in materials, and an afternoon in the shop. These plans are free — build one for your kid.

WHAT'S INSIDE

- Overview & Sizing Notes
- Materials & Hardware List
- Cut List & Layout Diagram
- Step-by-Step Assembly
- Padded Top Construction
- Safety, Finishing & Usage

Project Overview

This bench supports floor sitting work and pelvic positioning for children who don't need back support but benefit from a firm, height-adjustable surface. Slot channels allow tool-free height adjustment in 1" increments. The front and rear bolt columns are independent — set them to different heights to tilt the surface forward or backward. A removable padded top with a washable slip cover makes it comfortable and personalizable.

Small Model

This guide covers the Small size: 20"×10" top, 5-9" height range. The design is modular — build larger sizes using taller Outer Side panels. The top surface, hardware, and assembly steps stay the same.

20 × 10"

TOP SURFACE

5–9"

HEIGHT RANGE

1" Steps

ADJUSTMENT

3/4"

BAL TIC BIRCH

~\$50

MATERIALS EST.

Materials & Hardware

Wood

QTY	ITEM
1	2'×2' sheet — 3/4" Baltic birch plywood
1	~12"×22" scrap — 3/4" ply (padded top base)

Foam & Fabric

QTY	ITEM
1	1" upholstery foam, cut to 20"×10"
1 yd	Vinyl fabric for base cover wrap
—	Fabric for slip cover (child picks the pattern)
—	Upholstery staples + staple gun
—	Spray adhesive (foam to plywood)

Finish

QTY	ITEM
1 qt	Water-based polyurethane, satin
—	80, 120, and 220 grit sandpaper

Hardware

QTY	ITEM
4	1/4"-20 Star Knobs — black
4	1/4"-20 × 2" Hex Bolts
4	1/4" Flat Washers
4	1/4" Lock Washers
16	1-1/2" Wood Screws

Tools Required

- Circular saw — straight cuts
- Jigsaw — taper cuts and slot channels
- Plunge router + 1/4" straight bit — preferred for slots
- Drill + 1/4" bit + countersink bit
- Router + 1/4" roundover bit — all edges
- Upholstery staple gun
- Clamps, tape measure, square

ROUTER JIG AVAILABLE

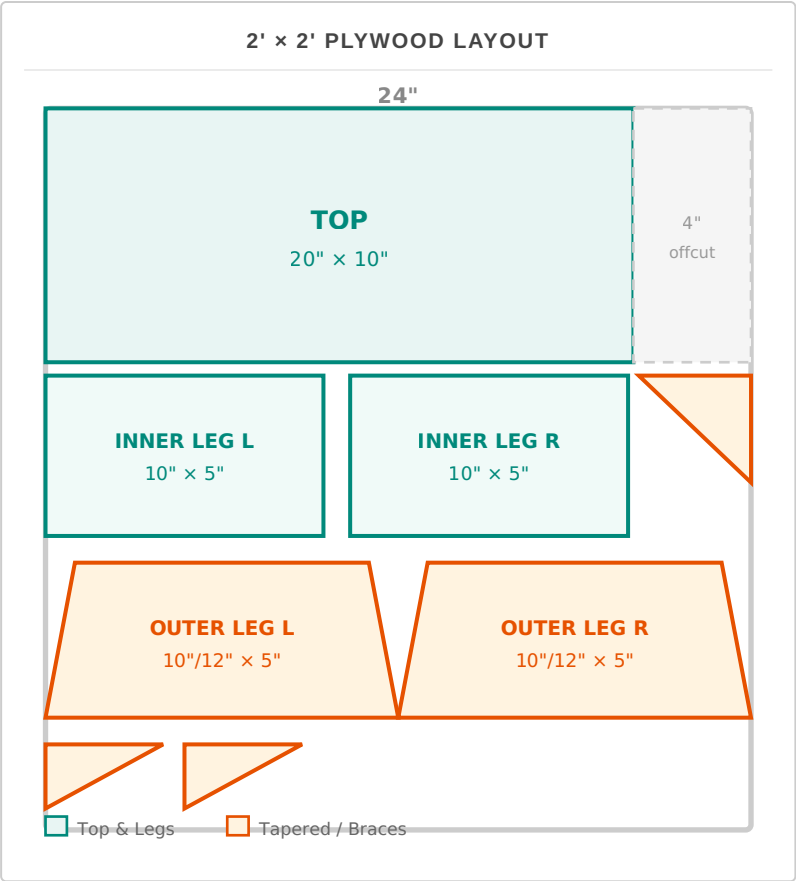
A free 3D printed jig for the slot channels is available as a free download — search "Built For Them router jig". Clamp to the leg panel, guide the router — consistent slots every time.

Cut List & Layout

PART	QTY	SIZE
Top	1	20" × 10"
Inner Leg	2	10" × 5"
Outer Leg	2	10"/12" × 5" tapered
Corner Brace	4	4" × 5" triangle

TAPER NOTE

Outer Legs are symmetric trapezoids: 10" wide at the top, 12" wide at the base — 1" of taper on each side over the 5" height. Both sides angle outward equally. Stack both blanks and cut together so they match exactly.



Step-by-Step Assembly

- 1

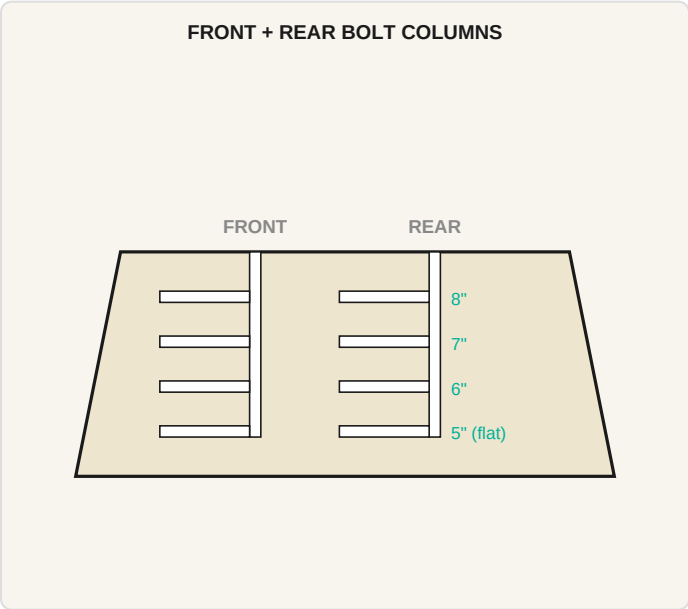
Mark All Parts on the Sheet
Lay out all parts on your 2'x2' sheet following the diagram. Label each part in pencil. Mark taper lines on both Outer Legs before cutting anything.
- 2

Make the Straight Cuts First
Cut the Top (20"x10") and both Inner Legs (10"x5") using your circular saw and a clamped straightedge. These establish all reference edges.
- 3

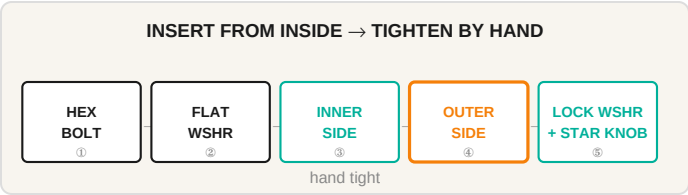
Cut the Outer Leg Tapers
Stack both Outer Leg blanks and clamp. Each piece is a symmetric trapezoid — 10" wide at the top, 12" wide at the base, with 1" of taper on each side over the 5" height. Mark both taper lines before cutting. Cut both pieces through the stack with a jigsaw so they match exactly.
- 4

Cut the Slot Channels in the Outer Legs
Two columns per leg, 4" apart center-to-center. Five slots per column, each 2" wide (horizontal), spaced 1" apart center-to-center vertically — giving height positions at 5", 6", 7", 8", and 9". **Router (preferred):** use the Built For Them 3D printed jig — clamp to the top edge, rout each slot in one pass. **Jigsaw:** mark both slot edges, drill a starter hole inside, cut and clean up with a chisel.

Outer Leg — Slot Layout



Hardware Assembly



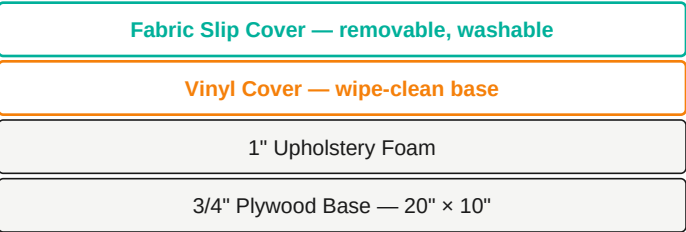
Tilt Feature

Front and rear bolt columns are independent. Set them to **different heights** to tilt the surface. Ask your child's therapist what angle works best.

Front	Rear	Effect
5"	5"	Flat — neutral
5"	6"	Gentle forward tilt
5"	7"	Moderate forward tilt
6"	5"	Backward tilt

Padded Top Construction

LAYER VIEW — 20" × 10"



Rests on bench top — no fasteners needed
Vinyl layer keeps foam clean between cover washes

- A** **Cut + Round the Plywood Base**
Cut 20"×10". Round all edges with 1/4" roundover bit. Sand to 120 grit.
- B** **Attach the Foam**
Cut foam to 20"×10" flush with the base. Secure with spray adhesive. Let cure before wrapping.
- C** **Wrap in Vinyl**
Add 3" all around. Pull firmly and staple to the underside. Start at centers, work outward. Miter corners neatly.
- D** **Sew the Slip Cover**
Fits over the padded top, open on the underside. Child picks the pattern. Vinyl base protects foam between washes.

Finished Build



What you're looking at:

- Slot channels cut into Outer Leg — 5 positions per column
- Two star knobs per side — hand-tight, tool-free adjustment
- Symmetric tapered Outer Legs — wider base for stability
- Removable padded top — 1" foam, vinyl base, slip cover
- Corner braces at all four inner corners

Completed v2 bench — slot channels, star knobs, and removable padded top with custom slip cover. Built from a single 2'×2' sheet of 3/4" Baltic birch plywood.

Safety, Finishing & Inspection

Edge Safety

Round every edge with a 1/4" roundover bit. Run your palm across all edges — anything sharp means rout or sand again before use.

Finishing

2 coats water-based polyurethane. Sand at 220 grit between coats. Wipe dust before final coat. Cure 24 hours before use.

Splinter Check

Wipe with damp cloth after sanding to raise grain. Let dry, re-sand at 120 grit until no grain raises on the wet test.

Hardware Check

After first use, re-tighten all 4 star knobs and confirm screws are fully seated. Re-check every few months and before every session.

Squareness Check

After attaching Inner Legs to the Top, set on a flat surface and check for rocking before adding corner braces.

Regular Inspection

Check before every session for splinters, cracks, or loose hardware. Do not use if damaged — this is load-bearing equipment.

⚠ **Sand and round all edges before use — child safety first.**

@BuiltForThemDIY · CC BY 4.0